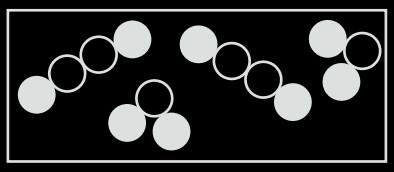
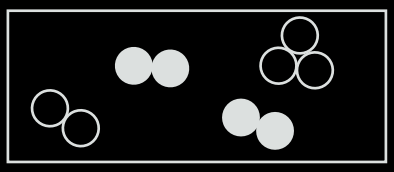
○ and ● represent atoms of different elements.

Which diagram shows a mixture of an element and a compound?

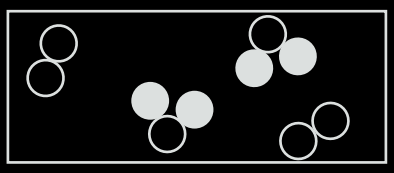
A



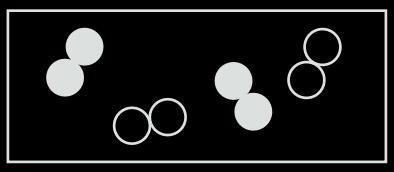
B



C



D



3 An aqueous solution of compound **X** reacts with aqueous sodium hydroxide to form a green precipitate and then aluminium powder is added. The mixture is heated and a gas that turns damp red litmus paper blue is given off.

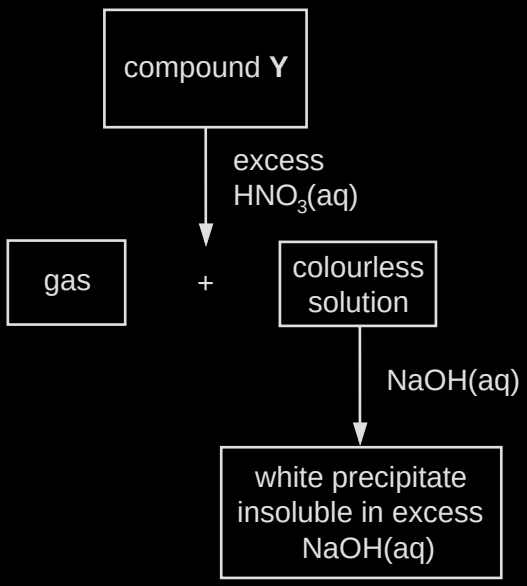
What is **X**?

A ammonium nitrate

B copper(II) chloride

C iron(II) nitrate

D iron(III) chloride



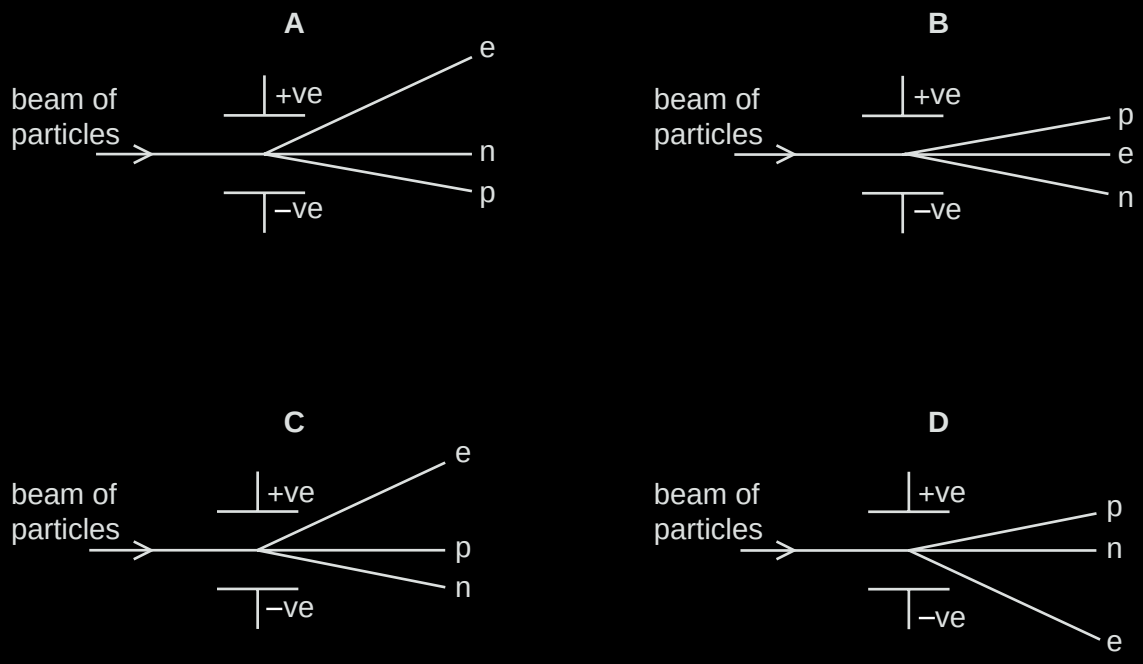
What could the compound **Y** be?

A aluminium sulphate

B calcium carbonate

C copper(II) carbonate

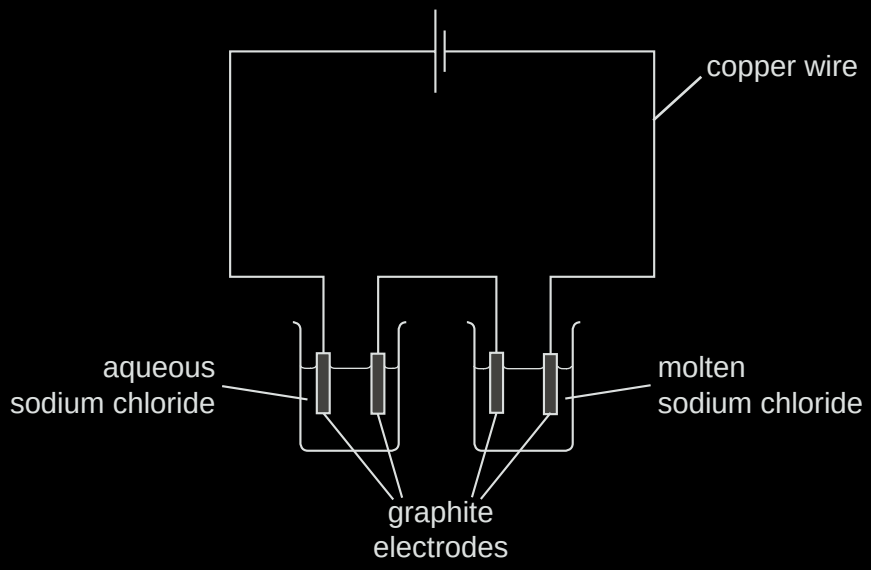
D zinc carbonate



7 The table shows the properties of some substances.

Which substance is a covalent compound?

	melting point / °C	electrical conductivity	
		of solid	of liquid
A	–38	conducts	conducts
B	–7	does not conduct	does not conduct
C	801	does not conduct	conducts
D	1540	conducts	conducts



Which substance has both positive ions and mobile electrons?

A aqueous sodium chloride

B copper wire

C graphite electrodes

D molten sodium chloride

9 Hydrogen can form both ionic and covalent compounds.

With which element will hydrogen form an ionic compound?

A carbon

B chlorine

C nitrogen

D sodium

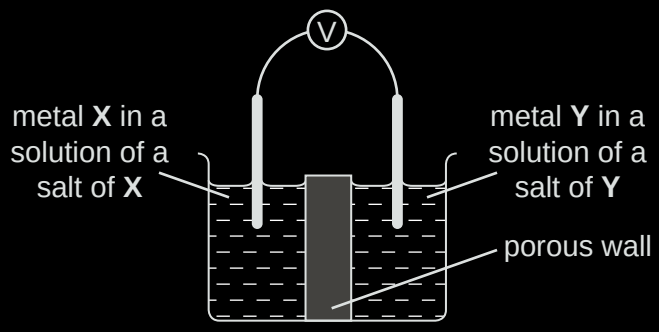
10 Which quantity is the same for one mole of ethanol and one mole of ethane?

A mass

B number of atoms

C number of molecules

D volume at r.t.p.



	metal **X**	metal **Y**
A	copper	silver
B	magnesium	silver
C	magnesium	zinc
D	zinc	copper

14 On combustion, which fuel **never** produces pollutants?

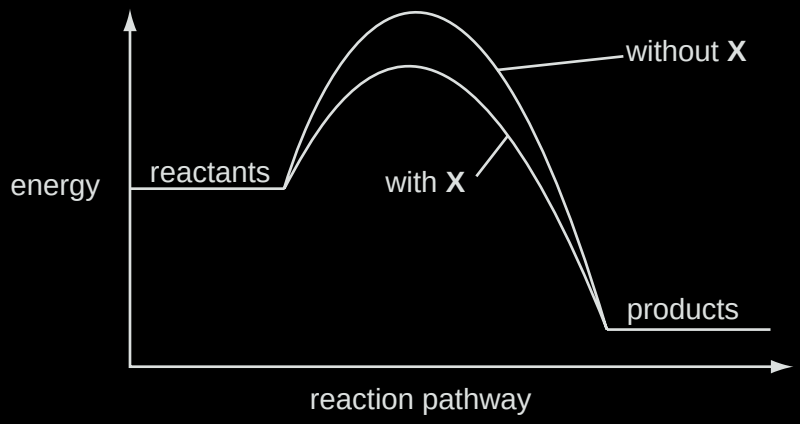
A diesel

B hydrogen

C methane

D petrol

$\rightleftharpoons$



Which change occurs when **X** is added to the reaction mixture?

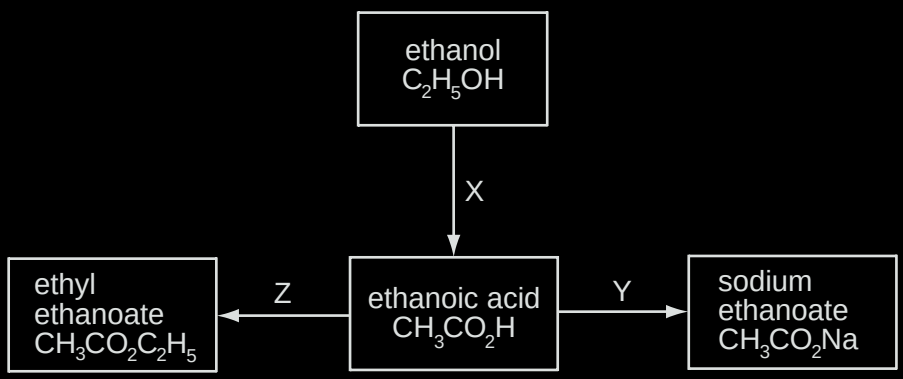
A The rate of reaction decreases.

B The rate of reaction increases.

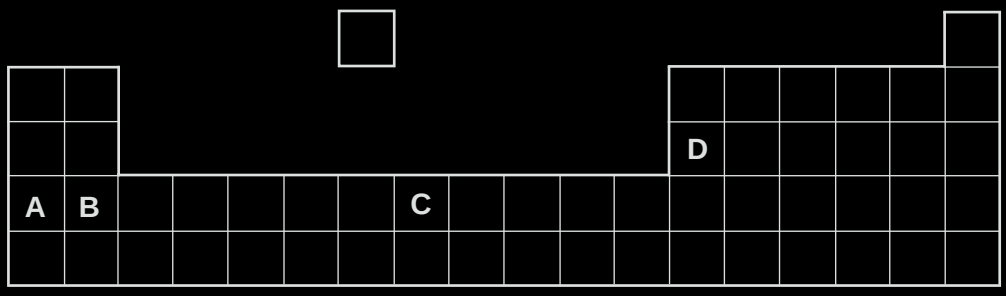
C The reaction becomes less exothermic.

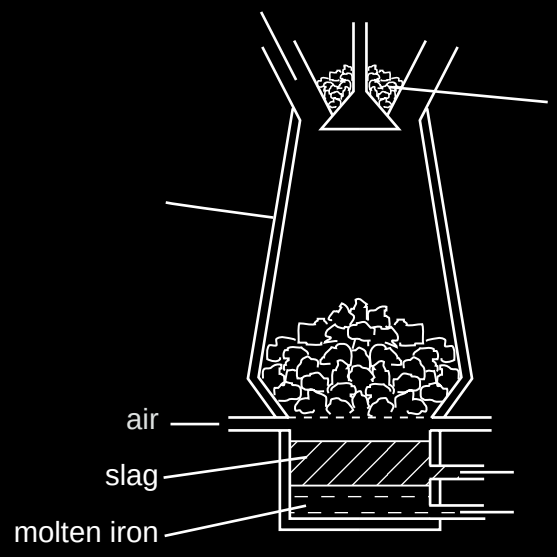
D The reaction becomes more exothermic.

17 Which of the reactions X, Y and Z involve oxidation?



A X only **B** X and Y **C** Y only **D** Y and Z


A
B
C
D



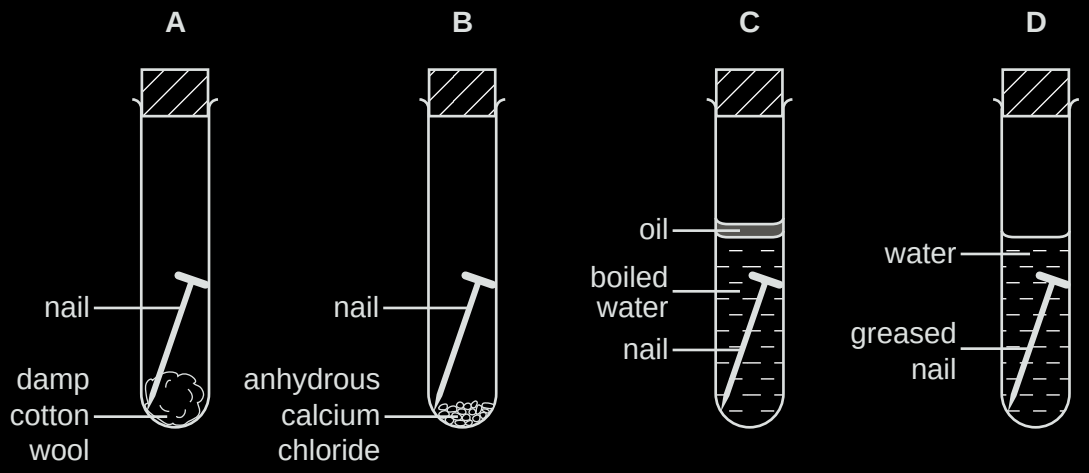
Which substance undergoes thermal decomposition?

A limestone

B carbon dioxide

C haematite

D slag



28 The carbonate of metal **X** is a white solid.

It decomposes when heated to form carbon dioxide and a yellow solid oxide.

What is metal **X**?

A copper

B iron

C lead

D sodium

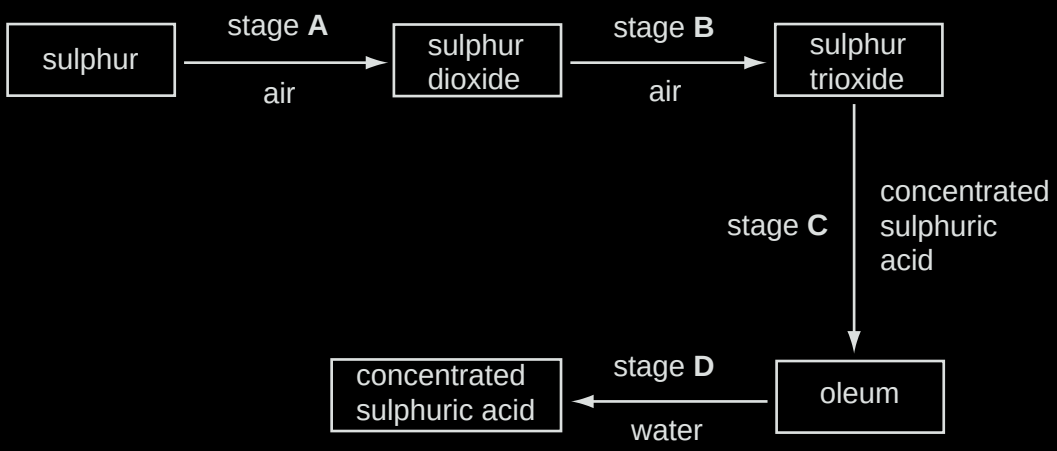
29 Which metal will displace hydrogen from aqueous solutions of acids but not from cold water?

A calcium

B copper

C sodium

D zinc


sulphur
stage A
air
sulphur dioxide
stage B
air
sulphur trioxide
stage C
concentrated sulphuric acid
oleum
stage D
water
concentrated sulphuric acid



What is the structural formula of **X**?

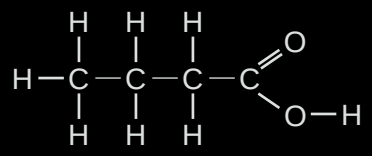
A $HCO_2CH_2CH_2CH_3$

B $CH_3CO_2CH_2CH_3$

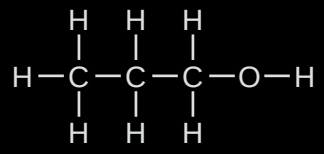
C $CH_3CH_2CO_2CH_3$

D $CH_3CH_2CH_2CO_2H$

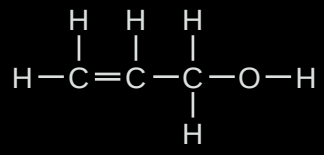
A



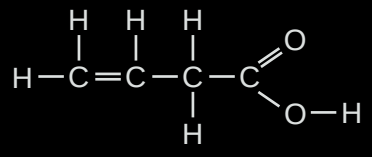
B

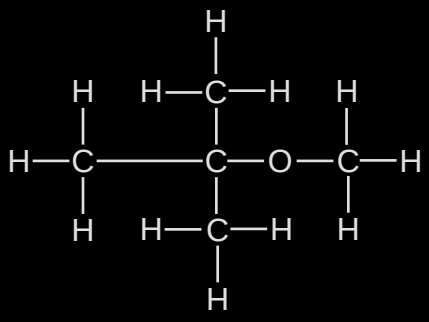


C



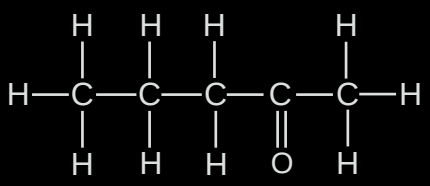
D



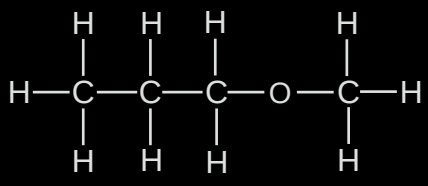


Which compound is an isomer of MTBE?

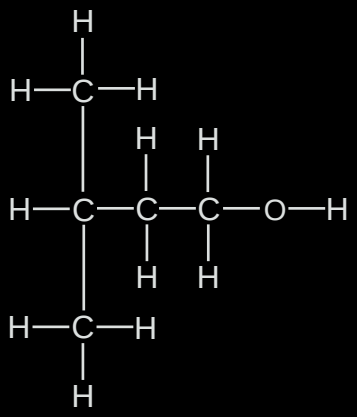
A



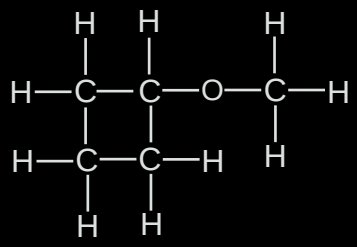
B



C



D



38 A liquid reacts with each of sodium carbonate, potassium hydroxide and ethanol.

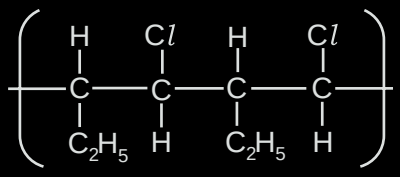
What is the liquid?

A aqueous ammonia

B ethanoic acid

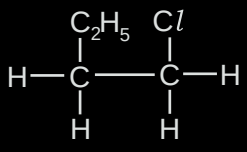
C ethyl ethanoate

D hydrochloric acid

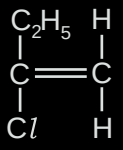


Which one of the following will form this polymer?

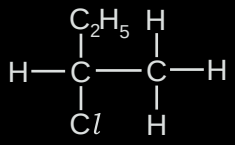
A



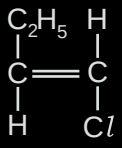
B



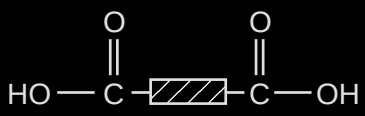
C



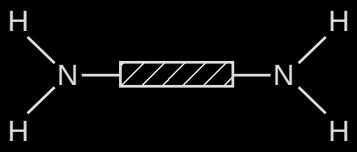
D



40 A polymer **X** was hydrolysed and the two products were



and



What can be deduced about **X**?

A It was a condensation polymer.

B It was starch.

C It was made by addition polymerisation.

D It was *Terylene*.

DATA SHEET
The Periodic Table of the Elements

Group																
I	II										III	IV	V	VI	VII	0
							1 H Hydrogen 1									4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4										11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12										27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulphur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	64 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
85 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57 *	178 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	Po Polonium 84	At Astatine 85	Rn Radon 86
Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89 †														

140 Ce Cerium 58	141 Pr Praseodymium 59	144 Nd Neodymium 60	Pm Promethium 61	150 Sm Samarium 62	152 Eu Europium 63	157 Gd Gadolinium 64	159 Tb Terbium 65	162 Dy Dysprosium 66	165 Ho Holmium 67	167 Er Erbium 68	169 Tm Thulium 69	173 Yb Ytterbium 70	175 Lu Lutetium 71
232 Th Thorium 90	Pa Protactinium 91	238 U Uranium 92	Np Neptunium 93	Pu Plutonium 94	Am Americium 95	Cm Curium 96	Bk Berkelium 97	Cf Californium 98	Es Einsteinium 99	Fm Fermium 100	Md Mendelevium 101	No Nobelium 102	Lr Lawrencium 103

*58-71 Lanthanoid series
†90-103 Actinoid series

Key

a
X
b

a = relative atomic mass
X = atomic symbol
b = proton (atomic) number

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).